

Band relation

Model — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Package

- **Completeness.** For all $p, q \in \Delta(X)$: $p \succcurlyeq q$ or $q \succcurlyeq p$.
- **Acyclicity.** There is no finite cycle of strict preferences $p_1 \succ p_2 \succ \dots \succ p_n \succ p_1$.
- **Quasi-transitivity.** Strict preference is transitive: if $p \succ q$ and $q \succ r$ then $p \succ r$.

0.2 Construction

Fix $u : X \rightarrow \mathbb{R}$ non-constant and $\varepsilon > 0$. Define an asymmetric relation P by $p P q$ iff $EU_u(p) - EU_u(q) \in (\varepsilon, 2\varepsilon]$, and let $p \succcurlyeq q$ iff not $q P p$. Then $\succcurlyeq$ is complete (P is asymmetric) and $\succ = P$. P is acyclic because EU_u strictly increases along any P -chain. But P is not transitive: with $EU_u(p), EU_u(q), EU_u(r) = 3\varepsilon, 1.5\varepsilon, 0$ we have $p P q$ and $q P r$ while $EU_u(p) - EU_u(r) = 3\varepsilon \notin (\varepsilon, 2\varepsilon]$.

0.3 Notes

Sanity-checked numerically in checks/countermodels.py. Not yet checked by a human.

Record: topics/decision-theory/models/band-relation.yaml